DATATHON 2026 · SWISS PROPERTY SEARCH

Robin real-estate search

Find your perfect Swiss home with natural language — from “bright flat near ETH flat near ETH with a balcony” to ranked results in seconds.

SYSTEM ARCHITECTURE

Seven stages from a sentence to a ranked list.

01 · INPUT
User Query
"bright flat near ETH, under 3k"

→

02 · LLM
Hard Fact Extraction Extraction
rooms, price, city, canton, radius

→

03 · DB
SQL Hard Filter
SQLite + haversine radius

→

04 · LLM
Soft Fact Extraction
brightness, modernity, family-friendly, value

→

05 · RANK
Multi-Signal Ranking
6 weighted signals, dynamic weights

→

06 · LLM
Claude Re-Ranker
top-30 reasoned by Claude Sonnet

→

07 · OUTPUT
Results UI
chat + list + map, explanations

LLM STEP STRUCTURED DATA RETRIEVAL / UI

THE RANKING ENGINE

Six signals, one score - weights adapt to the query.

SIGNAL 01

BM25 · Text

Lexical match against listing title + description, the reliable reliable baseline.

SIGNAL 02

BM25 · Captions

Search inside AI-generated captions of every listing image. image.

SIGNAL 03

Image Embeddings

Jina CLIP v2 cosine similarity between query text and listing listing photos.

SIGNAL 04

Feature Match

Balcony, parking, pets, elevator — structured attribute overlap. overlap.

SIGNAL 05

Geo / Distance

Proximity to public transport, schools, kindergartens — the the commute score.

SIGNAL 06

Landmark Proximity

Geocoded POIs → haversine distance when a named place is in place is in the query.

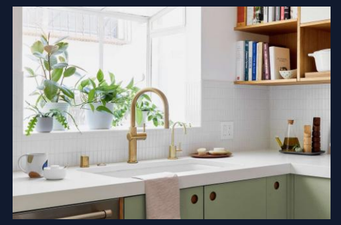
Top-30 pre-scored → re-ranked by **Claude Sonnet** for soft-signal reasoning → **Final** ordered list

IMAGE INTELLIGENCE

Two image pipelines, one visual understanding.

PIPELINE A

AI-Generated Captions



→ `claude-haiku` caption every listing image

→ `index` BM25 text search over captions

> "bright kitchen with marble countertops, windows"

Visual content becomes searchable text.

PIPELINE B

Semantic Embeddings

→ `jina-clip-v2` encode all images offline

→ `cosine` similarity vs query text at runtime

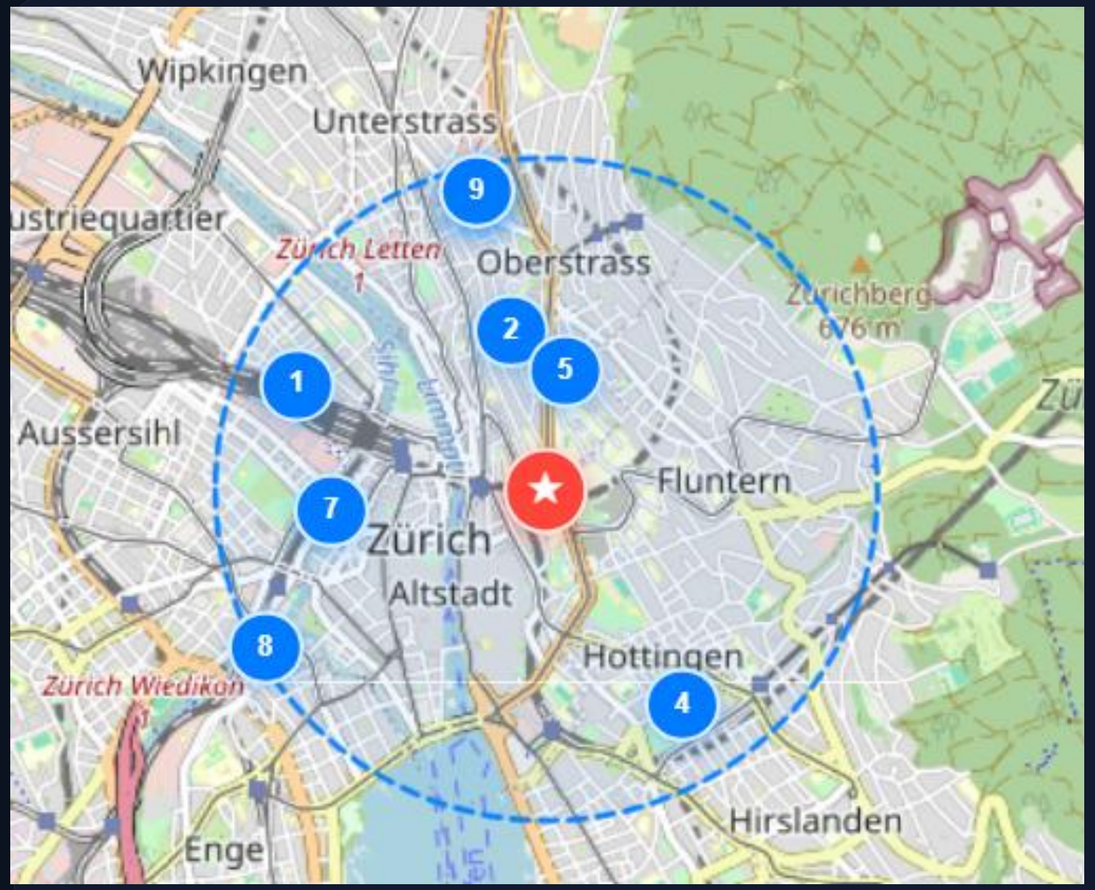
```
[ 0.023, -0.187, 0.442, 0.091, -0.318, 0.276, … 1024 dims ]
```

Visual content becomes a vector.

NET EFFECT Query "sunny apartment with modern kitchen" finds visually matching listings — even when the description never says

LOCATION & LANDMARKS

Landmark proximity – hard vs soft filter based on query



USER SAYS

"bright apartment within 1.5 km of ETH Zürich"

01 → LLM extracts landmark: "ETH Zürich"

02 → Geocode → 47.3764°N, 8.5485°E

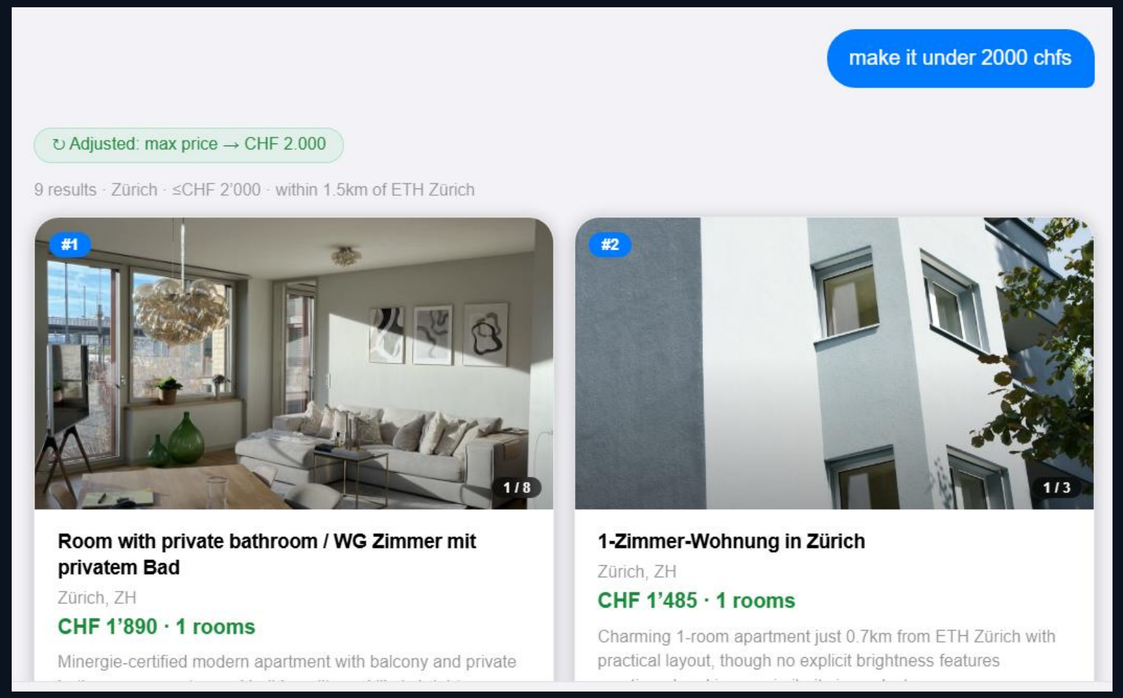
03 → Haversine radius filter: r ≤ 1.5 km

04 → Proximity score feeds ranker (landmark_w = 0.30)

05 → Results pinned on Leaflet + OSM map

CONVERSATIONAL UI

One chat, multi-turn memory, maps & gallery built in.



- **Multi-turn memory**

  Follow-ups like "make it cheaper" carry carry forward every constraint you set set before.

LIVE DEMO

Ask anything.

```
POST /listings { "query": "…" }
```

"3-room in Zürich under 2'800 with a balcony"

"sunny Apartment near a kindergarten in Basel"

"1.5 km from Hauptbahnhof, pet-friendly, modern kitchen"